Attention Research

Results across two projects

PROJECT 1

CRPA: is behavioural contribution measurable at all?

Audited a partitioned-attention repository, then tested whether its own central claim survives a resolvability check.

1.1 The claim that was withdrawn

The branch reported that structural overlap does not predict behavioural contribution. Check 0 measures each edge's effect twice on disjoint halves of the evaluation split and correlates the two estimates. **Pooled verdict: UNRESOLVABLE.**

seed	r_delta	r_stat	ceiling	observed rho	delta size	converges
42	+0.088	+0.119	0.102	+0.018	6.0 ULP	no
1337	-0.026	+0.255	0.000	-0.017	5.0 ULP	no
2024	+0.012	+0.250	0.054	+0.013	4.0 ULP	no

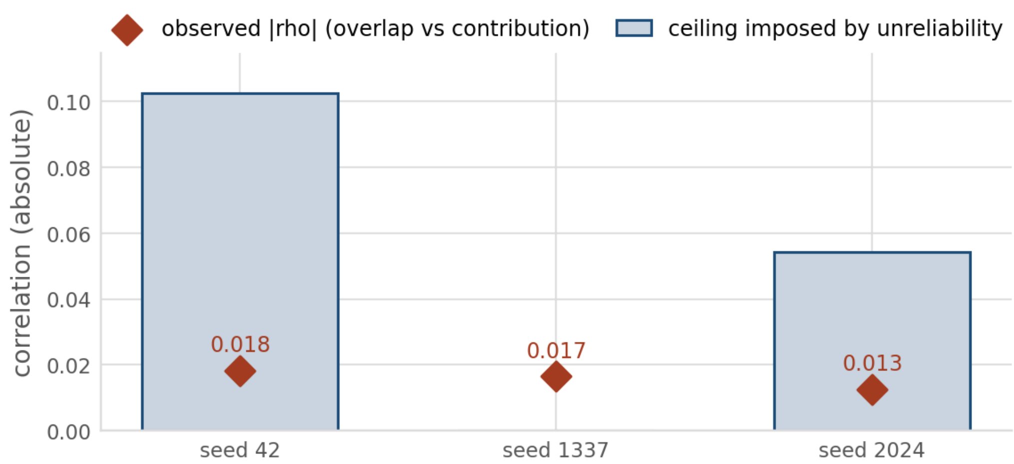

Figure 1. Observed correlation against the ceiling that measurement unreliability imposes on it, sqrt(r_delta × r_stat). Every observed value sits far inside its own ceiling, so a real decoupling and measured noise cannot be told apart. Seed 1337 has no bar: its split-half reliability is negative, so the ceiling degenerates to zero and that seed carries no information at all. n = 3 seeds, 1,044 to 1,152 scored edges each.

1.2 The floor holds across three orders of magnitude

The replacement claim is that single-edge contribution is not resolvable, not that overlap fails to predict it.

scale	precision	single-edge delta
12.4M (Tier 1)	float32	4 to 6 ULP, split-half reliability 0.088
138M (Tier 2)	float32	single-edge p90 0 ULP; group of 24 is 1 ULP
6.9B (Tier 3)	**bfloat16**	**exactly 0.000e+00 across 192 edges**

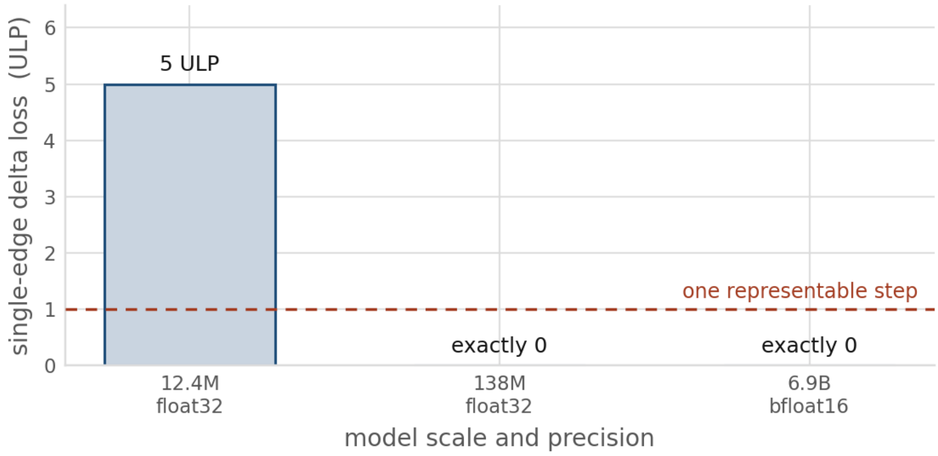

Figure 2. Typical single-edge intervention effect, expressed in units of the least representable step at the working precision. At 138M the single-edge p90 is zero; removing all 24 candidates together moves the loss by one ULP. At 6.9B the single-edge effect is exactly zero: the edit shifts logits by 2.58, 3.05 and 0.127 at layers 0, 16 and 31, and the loss is bit-identical to baseline for all 192 edges. A criterion thresholding this quantity is unfalsifiable at every scale tested.

1.3 Reproduction of the published table

Re-run from the original commit. Four of five rows come back. The one that does not is the one the paper's claim rests on.

Variant	Published	Reproduced	Verdict
Dense Transformer	50.9%	55.6%	reproduces
Sliding Window	51.9%	52.5%	reproduces
CRPA no regularisation	8.4%	10.9%	reproduces
CRPA naive regularisation	5.3%	5.3%	reproduces exactly
CRPA causal regularisation	**32.8%**	**7.5%**	**does not reproduce**

1.4 Three seeds against a measured chance floor

The floor is measured by simulating the task generator, not asserted from vocabulary size: 52.78% (sd 0.08, n = 25,000 x 5). **No variant clears it, dense included.** The retrieval task cannot discriminate between these variants.

Variant	Retrieval	t vs floor	Verdict	Overlap
dense	53.6%	+0.8	indistinguishable	0.403
sliding window	47.2%	-1.4	indistinguishable	0.322
crpa_noreg	4.4%	-66	below	0.244
crpa_naive	4.6%	-100	below	0.264
crpa_contribution	4.3%	-50	below	0.222

1.5 Long context: the wall was in the diagnostic

A fused dense kernel is faster than the sparse gather path at every length benchmarked, with the gap narrowing as context grows. The benchmark itself was not run beyond 16k.

Context	Dense	Sliding	CRPA	CRPA vs dense	Peak memory
4,096	13.2 ms	19.5 ms	31.5 ms	2.40x slower	737 MB
8,192	30.0 ms	62.4 ms	60.1 ms	2.00x slower	1,181 MB
16,384	72.0 ms	218.1 ms	119.1 ms	1.65x slower	2,067 MB

1.6 32k and 64k, previously unreachable, now measured

Six attempts on 48GB and 80GB cards had all exhausted memory. The cost was never in the model: bounding the candidate-edge diagnostic closed both lengths on the same hardware.

Context	Peak before	Peak after	Retrieval	Realized overlap	Largest single-edge delta
4,096	14.14 GB	0.75 GB	0.0%	0.2265	9.54e-07
8,192	27.74 GB	1.15 GB	0.0%	0.2256	9.54e-07
16,384	55.04 GB	1.24 GB	0.0%	0.2297	1.91e-06
32,768	*out of memory*	1.93 GB	0.0%	0.2308	9.54e-07
65,536	*out of memory*	3.30 GB	0.0%	0.2302	9.54e-07

The measurement extends the central claim rather than complicating it. The largest single-edge delta is **9.5367e-07 at four of the five lengths, exactly one unit in the last place**, and two at 16,384, against a float32 resolution of 1.32e-06. Single-edge behavioural contribution sits at the measurability floor across a sixteenfold range of context, not only at the short lengths where it was first seen. All five rows come from one implementation at one chunk size, so the memory column compares like with like.

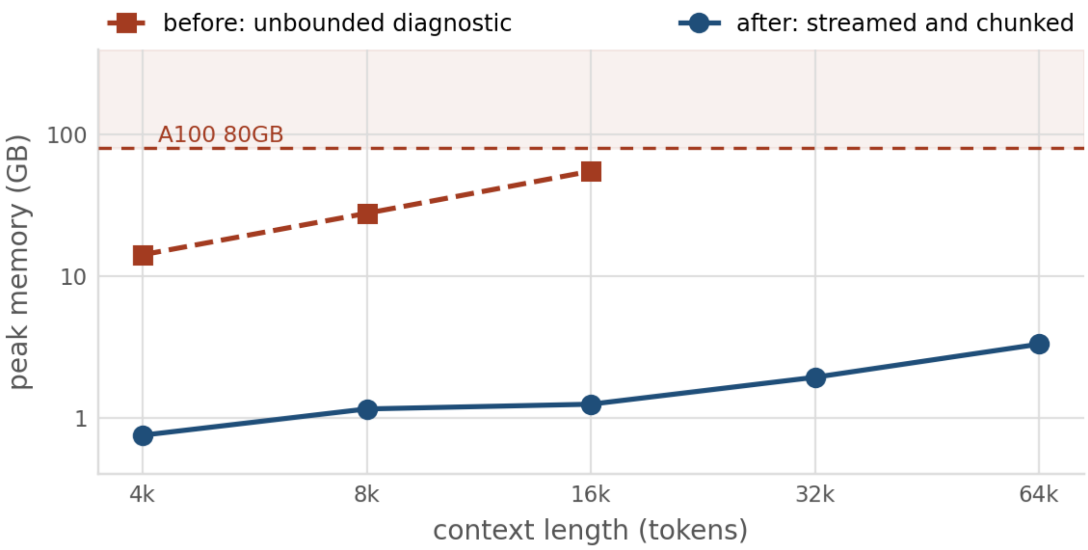

Figure 3. Peak memory of the candidate-edge diagnostic before and after bounding it. The unbounded implementation grew at 3.45 GB per 1,000 tokens and a linear fit to its three measured points predicts 110 GB at 32k and 219 GB at 64k, both beyond the 80GB card. Streaming the overlap measurement by layer, disabling autograd in the group diagnostic and scoring edges in small chunks flattens the profile to 0.75-3.30 GB across the whole range, a 44x reduction at 16k. n = 1 seed, 138M parameters, bfloat16.

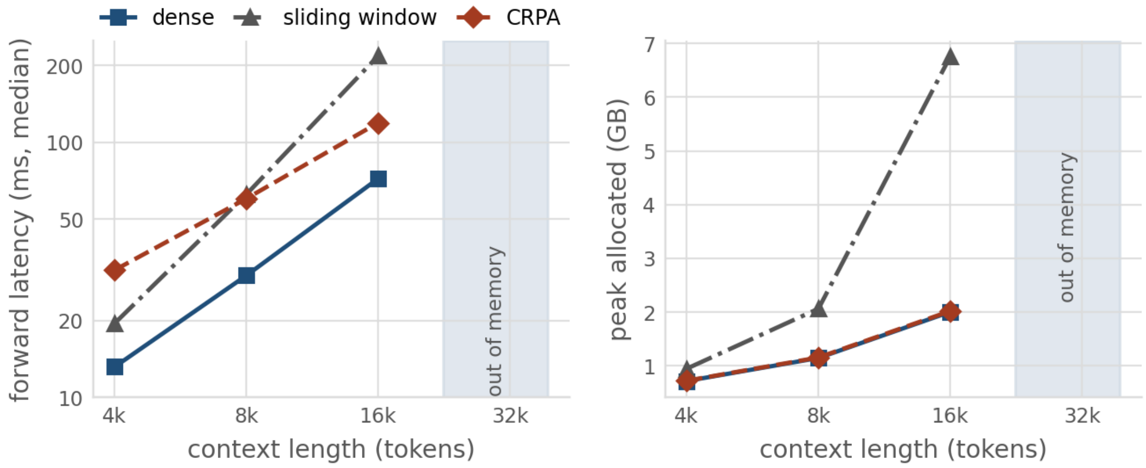

Figure 4. Measured forward latency and peak allocated memory at the three lengths the latency benchmark was run at. A fused dense kernel is faster than the sparse gather path at every one, and the gap narrows from 2.40x to 1.65x as context grows, so the sparse path is closing but has not overtaken it by 16k. The shaded region marks lengths not benchmarked for latency; the diagnostic does now run there, as section 1.6 reports. Batch size 1, bfloat16, 138M parameters, median of 10 iterations after 5 warmups.

1.7 Defects found and repaired

Each of these changed a reported number.

Defect	Consequence
Central claim unsupported by its own measurement	A decoupling claim the data cannot support. Withdrawn
Chance floor asserted, not measured	Real floor 52.78%, not 5.0%. Every above-chance verdict inverted
Intervention masked a query-row pair, not a query-to-key edge	The measured object was not the object being reasoned about
Roughly half of all interventions were causally inert	Delta zero by construction, so the pair was filed redundant
Short runs recordable as completed measurements	Nine 3-iteration runs entered an aggregate and moved it by 26x
Auxiliary routing entropy pinned at ln(4)	A constant reported as a finding

xsa-controls: sanity checks for attention surgery

Three checks shipped as importable code: is the measurement resolvable, does the statistic beat an anisotropy null, and does a matched arbitrary intervention recover the gain?

2.1 Check 1 across a nine-model scale ladder

5,408 head-level rows, 32 real documents per model, eager attention, null partner drawn within the sequence from positions the query could causally attend.

model	params	cos_self	cos_null	excess	% self-specific
gpt2	124M	0.4828	0.2987	0.1840	38.1
pythia-160m	160M	0.4180	0.2637	0.1544	36.9
gpt2-medium	355M	0.4252	0.2579	0.1674	39.4
pythia-410m	410M	0.4022	0.1937	0.2086	51.9
gpt2-large	774M	0.4213	0.2117	0.2096	49.7
pythia-1.4b	1.4B	0.3862	0.1900	0.1963	50.8
gpt2-xl	1.5B	0.3861	0.2069	0.1792	46.4
pythia-2.8b	2.8B	0.3565	0.1859	0.1705	47.8
pythia-6.9b	**6.9B**	**0.3404**	**0.1979**	**0.1425**	**41.9**

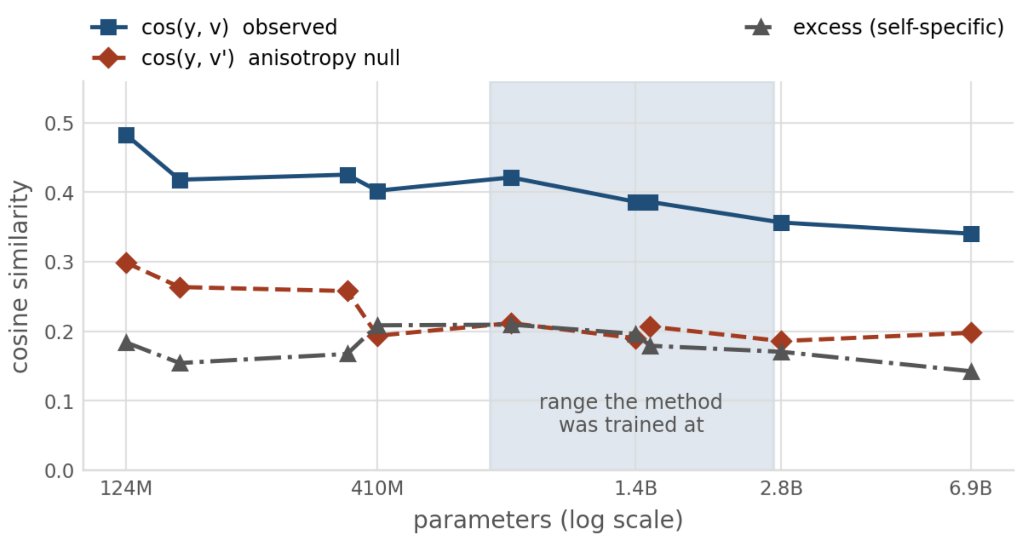

Figure 5. The anisotropy null weakens with scale but does not vanish. cos_null falls from 0.2637 at 160M to 0.1979 at 6.9B, yet 58% of the raw statistic is still explained by the null at 6.9B, and about half across the shaded range where the method was actually trained. This answers the strongest objection to the framing, that large models are isotropic, with measurement rather than extrapolation. n = 9 models, 5,408 heads.

2.2 The null depends on where it is measured

The same model and corpus, measured at five context lengths.

Context length	64	128	256	512	1024
cos(y, v) observed	0.4900	0.4846	0.4812	0.4772	0.4743
cos(y, v') null	0.3747	0.3361	0.3096	0.2904	0.2788
self-specific share	**23.5%**	**30.7%**	**35.7%**	**39.2%**	**41.2%**

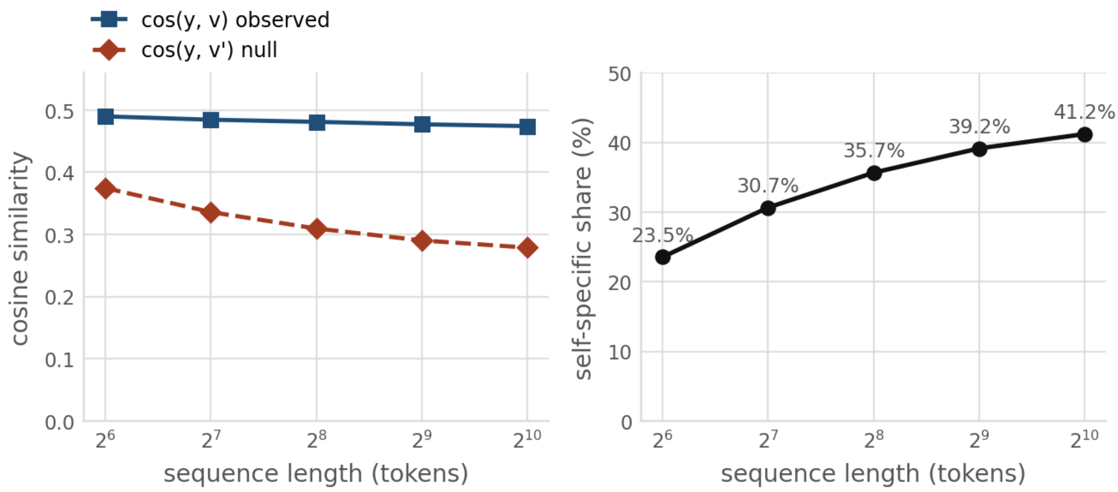

Figure 6. The observed statistic is nearly flat across a sixteenfold change in context length while the null moves by a third, so the self-specific share swings from 23.5% to 41.2%. A claim of the form “only N% of this statistic is self-specific” is therefore a property of the measurement context as much as of the model, and Check 1 must always be reported with the length it was measured at. n = 24 documents per length, GPT-2.

2.3 Grouped-query attention, which nobody had checked

Under GQA several query heads share one KV head, so the token's own value vector is shared across a group. The source method's own table reports no KV-head count.

model	query / KV heads	within-group excess	across-group excess
Qwen2.5-0.5B	14 / 2	**+0.2415**	**-0.1876**
Qwen2.5-1.5B	12 / 2	**+0.2731**	**-0.1922**
TinyLlama-1.1B	32 / 4	**+0.2373**	**-0.1126**

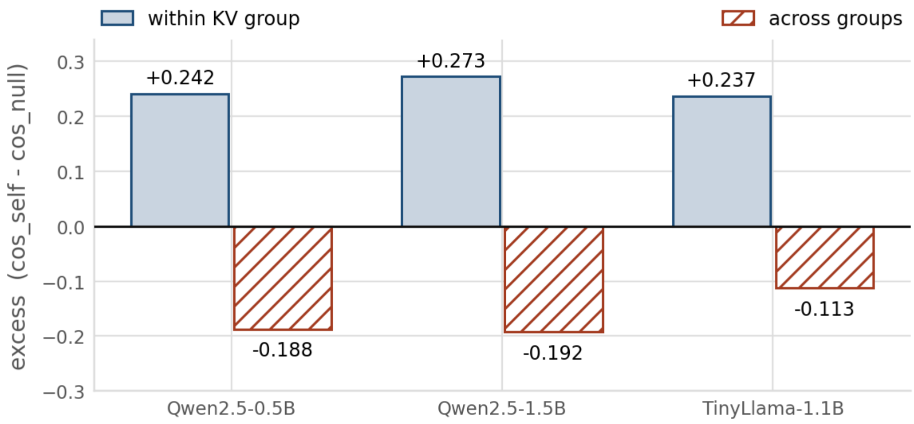

Figure 7. Borrowing a neighbouring KV group's value at the same position does not merely lose the effect, it reverses it. The self-value direction is specific to the head's own group. GQA models also show a higher self-specific share, 56 to 68%, than any multi-head model on the ladder, 37 to 52%, so grouped-query attention is not a neutral change of variable for this family of methods. The sign replicates outside the Qwen family: TinyLlama-1.1B is a Llama architecture with a different group ratio and its across-group excess is also negative. The magnitude is about 40% smaller, so the sign generalises and the size does not. n = 3 models across 2 families, 1,376 heads.

2.4 Does the statistic predict its own intervention?

Each head's self-value is removed in a frozen model and the change in loss measured twice, on disjoint halves of the evaluation documents. Nothing is reported until A × V rebuilds the model's own attention output.

model	statistic	raw rho	split-half r	ceiling	disattenuated	verdict
gpt2	cos(y, v)	**+0.462**	+0.795	0.890	**+0.519**	reliable
gpt2	excess	+0.279	+0.795	0.891	+0.313	reliable
pythia-160m	cos(y, v)	+0.149	+0.419	0.645	+0.231	attenuated
pythia-160m	excess	+0.189	+0.419	0.646	+0.292	attenuated
pythia-410m	cos(y, v)	**-0.025**	+0.531	0.726	-0.034	attenuated
pythia-410m	excess	**+0.249**	+0.531	0.727	**+0.342**	attenuated

The three models disagree, and the disagreement is the result. In **GPT-2** the raw self-value cosine predicts (+0.462) and the excess much less (+0.279). In **pythia-410m** the ordering reverses completely: the raw cosine carries **nothing** (-0.025) while the excess carries some (+0.249). **pythia-160m supports no ordering either way**, both being close and weak (+0.149 against +0.189), and is reported as abstaining rather than as weak evidence for one side.

One model each way and one abstaining supports the narrow claim — whether a raw statistic predicts its own intervention is model-dependent — and does not support the stronger reading that the null correction rescues prediction everywhere.

Each statistic is disattenuated by **its own** split-half reliability, and both halves of the statistic are pooled to match the pooled effect. An earlier revision did neither, and the difference was not cosmetic: pythia-160m's excess correlation fell from +0.487 to +0.189 once the pairing was made symmetric.

2.4.1 The first version of that table did not reproduce

The same experiment was run three times: twice at 24 evaluation documents per half and once at 64, on two different GPUs under two different PyTorch and transformers versions.

quantity	run 1 (n=24)	run 2 (n=24)	run 3 (n=64)	verdict moved?
gpt2 r_delta	+0.752	+0.629	**+0.795**	no, reliable throughout
pythia-160m r_delta	+0.304	+0.194	**+0.419**	**yes**
pythia-410m r_delta	+0.446	-0.007	**+0.531**	**yes**
gpt2 cos(y,v) *disattenuated*	+0.521	+0.527	**+0.519**	no

At 24 documents per half the reliability estimate for the two Pythia models is **not stable**. Repeating the run moved pythia-410m's from +0.446 to -0.007, taking the verdict from attenuated to unresolvable, and the correlations followed it down. Any statement built on the first run's Pythia numbers would not have survived a reviewer repeating the measurement.

At 64 documents per half both models are stable, +0.473 and +0.435, and the pattern the first run showed returns. That is the run reported above. Both smaller runs are kept in the repository rather than discarded, because the disagreement is more informative than either alone.

The last row is the argument for disattenuation, and it makes that argument better than any reasoning could. GPT-2's raw correlation moved across these runs, from +0.450 to +0.469. Its disattenuated value did not: +0.521, +0.527, +0.526, +0.519. Disattenuation divides out precisely the reliability that moved, so what is left is stable across budgets, GPUs and library versions.

This is Check 0 catching a claim in our own work, for the second time in this project. An effect that is not reliably measurable produced a correlation that did not survive being measured again.

2.4.2 Why this disagrees with the prior figure by 10x

The specification quotes prior GPT-2 values for exactly this correlation. Our GPT-2 number is an order of magnitude larger, so the gap has to be explained rather than left for a reader to find.

statistic	prior value	measured here	ratio
cos(y, v)	0.043	**+0.462**	10.7x
excess	0.017	+0.279	16.4x
a_ii	-0.021	*not measured*	–

The prior values were measured on **one paragraph repeated two hundred times** — base loss 0.76 nats against 3.96 for real prose. The specification's own bug list flags that input as a defect to fix before porting anything.

The reason it matters here is specific rather than general. **A correlation across heads needs variation across heads.** One paragraph repeated gives a model very little to do differently in different heads, so the per-head effects it produces are small and largely undifferentiated, and correlating them against anything returns approximately zero. Near-zero on that input is a property of the input, not of the method.

Measured on 64 real wikitext-103 documents per half, with disjoint halves and the reliability of the effect established first, the same correlation is **+0.462 with a ceiling of 0.890**. The prior figures are superseded rather than contradicted: they are not measurements of the quantity they appear to describe. Taken at face value they would have said the motivating statistic is unrelated to where the intervention helps, on every statistic and every model, which is the opposite of what GPT-2 shows and only half of what Pythia shows.

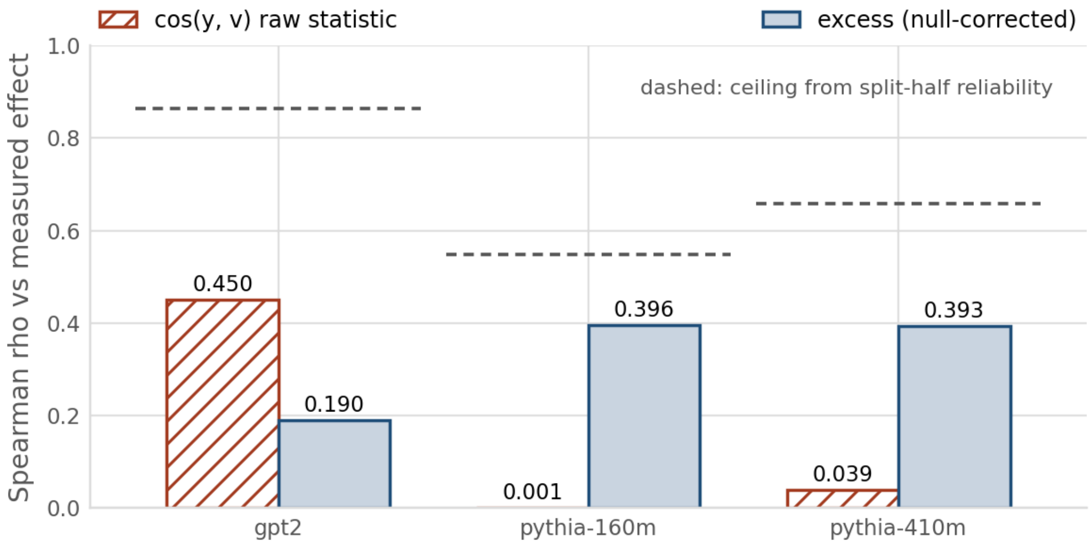

Figure 8. Correlation between each per-head statistic and the measured effect of removing that head's self-value. Dashed lines mark the ceiling that split-half unreliability places on any observable correlation, sqrt(r_delta × r_stat); every bar sits below its own ceiling, which is what makes the comparison legitimate. This is the check that returned UNRESOLVABLE on the sibling CRPA project: here the effect is resolvable, so the correlation can be read. n = 3 models, 144 to 384 heads each.

2.5 The matched intervention (underpowered pilot)

24 cells, 3 arms x 8 seeds, identical initialisation and data order per seed. Measured step-0 deviation across all five arms is 0.000e+00, so the arms start from a common point. This ran at 5e7 tokens per run, outside the pre-registered [3.5e8, 6e8] band, and is reported as the underpowered pilot; the primary endpoint is section 2.6.

arm	mean delta vs baseline	95% CI	t	p	n
random (primary)	**+0.001190**	[+0.000351, +0.002040]	+2.48	**0.042**	8
xsa	+0.001515	[-0.001223, +0.004807]	+0.92	0.387	8

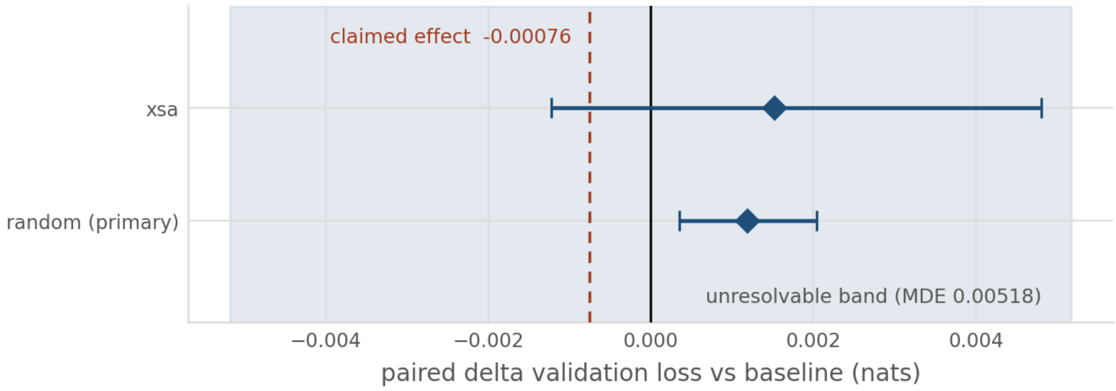

Figure 9. Both arms fall inside the shaded band of effects the design cannot resolve. The minimum detectable effect is 0.00518 nats against the 0.00076 the method's own independent replication reports, so the study is underpowered about sevenfold and cannot settle the question in either direction. The two arms are also indistinguishable from each other, which is what Check 2 asks. Reported as a power failure, not as a null result. n = 8 seeds per arm.

2.6 The primary endpoint, registered and running

The pilot above was void as a primary endpoint because it ran outside the pre-registered band. Re-running it properly was blocked by a cost projection that turned out to be measuring the wrong thing.

arm	measured throughput	hours for 8 runs	at $0.74/hr
baseline	176,467 tok/s	5.04	$3.73
xsa	159,358 tok/s	5.58	$4.13
random	166,267 tok/s	5.35	$3.96
total, 24 cells	–	**15.96**	**$11.81**

The committed calibration projected **$24.01** for this design, which scaled to the registered budget is about $27 and breaches a $20 ceiling. That figure is solved from **diagmask's** throughput of 71,918 tok/s, because diagmask is the slowest of the five arms and the solver sizes the whole design against its worst case. Diagmask is not in this run. Measured on the card that would run them, the three arms present are between 159,000 and 176,000 tok/s, roughly 2.3x faster, and the real projection is **$11.81**. Substituting the slowest arm's throughput for arms twice as fast would have blocked a run that fits with $8 to spare.

2.7 What was registered, and what was dropped

Registered in the budget ledger before the run started, not after it finished.

item	value
configuration	CFG_S, arms baseline / xsa / random
seeds	42, 1337, 2024, 7, 99, 512, 8191, 31337
tokens per run	399,900,672
inside pre-registered band	yes, [3.5e8, 6e8]
batch aligned	yes, 131,072 x 3,051
primary endpoint	random vs baseline; Holm over secondaries only
CFG_M scale check	*dropped for budget*

CFG_M is dropped, with the arithmetic rather than a shrug. At CFG_M the measured diagmask slowdown is 2.336 and the same calibration projects $51.57 for 24 runs at the lower budget, about $59 at the registered one. Even applying the same throughput correction as above, the floor sits well beyond $20. Dropping it follows the specification's own pre-registered priority order, which cuts the CFG_M scale check first and the secondary arms second. It is recorded as dropped for budget, which is its sanctioned status, not as an oversight.

2.8 The checklist discriminates between methods

Check 1 applied to two further methods on frozen GPT-2, each with a null matched to the structure the statistic inherits for free.

method	statistic	null	excess	% self-specific	survives
attention sinks	0.4028	0.0034	0.3994	99.2	yes
massive activations	12.8719	3.6452	9.2266	71.7	yes
self-value (this work, 6.9B)	0.3404	0.1979	0.1425	41.9	no

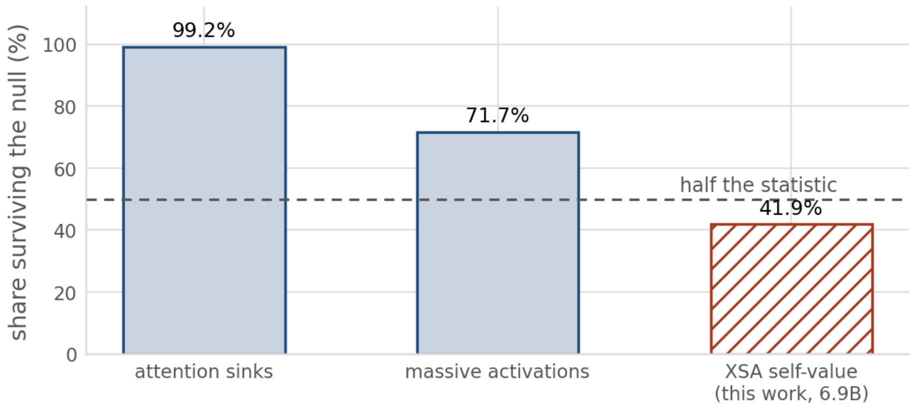

Figure 10. Attention sinks retain 99.2% of their statistic after a null matched for recency, and massive activations 71.7% against the maximum expected from a matched-variance Gaussian. The self-value statistic retains 41.9%. The checklist separates methods rather than debunking all of them, which is a stronger position than a blanket null.

2.9 A reproduction left failing on purpose

Published GPT-2 reference: cos_self 0.5406, cos_null 0.3798, excess 0.1608. Thirteen measurement conventions were tested and the full grid is reported unselected.

convention varied	range of cos_self observed	reproduces reference
sequence length, 64 to 1024	0.4827 to 0.4955	no
position 0 included	0.4847	no
null partner definition, 3 variants	0.4832	no
head pooling	0.4832	no
layer subsets, 4 variants	0.4405 to 0.6494	no

No tested convention reproduces the reference. Layer subsets move the statistic most, so a subset is the likeliest remaining explanation, but no tested subset lands on the reference triple. Reported as unexplained rather than resolved by further search: selecting the setting that hits a target and presenting it as the method is the practice this work argues against.

2.10 Defects found only by running it on GPU

These defects were not visible from CPU testing.

Defect	Consequence
diagmask crashed on every GPU run	bf16 autocast dtype mismatch. One of five arms could never have run
Calibration output was never consumed	The factorial used a default budget, so the gate changed nothing
Budget clamped up to an unaffordable floor	Returned a plan that could not be paid for and called it fine
The spend threshold was never enforced	Defined in config, referenced nowhere
Two runs mixed token budgets in one directory	Caught by a guard ported from the sibling project; data discarded

Status, and what remains open

Both deliverables carry green CI. No compute is running.

	CRPA	xsa-controls
Tests	266 passing	202 passing
Coverage	83%	90%
Figures from real data	7 of 7	5 of 5
Compute used	A6000, RTX 6000 Ada, A100 80GB	RTX 6000 Ada, A100 80GB
Open item	long context at 32k and 64k	GPT-2 reference reproduction
Why it is open	proven memory wall over five attempts	thirteen conventions tested; no convention matches